

Question:

A 31-year-old man with untreated HIV infection is admitted to the hospital because of a 3-day history of blurred vision and flashing lights in his left eye. Indirect ophthalmoscopy shows retinal hemorrhages of the left eye. Treatment with a drug that directly inhibits viral DNA polymerases by binding to pyrophosphate-binding sites is initiated. Two days later, the patient has a generalized tonic-clonic seizure. This patient's seizure was most likely caused by which of the following? Answer Choices: A. Demyelination B. Hypocalcemia C. Hypoglycemia D. Lactic acidosis

Reasoning + Answer:

<think> To answer this question, we need to look at the patient's symptoms and the possible side effects of the treatment he is receiving. The patient is being treated with a drug that inhibits viral DNA polymerases, which is a type of medication that is commonly used to treat HIV infection. One of the possible side effects of this type of medication is a condition called lactic acidosis, which can cause a range of symptoms including seizures. The patient's symptoms of generalized tonic-clonic seizure are consistent with the possible side effects of the medication, and the fact that he has been experiencing other symptoms such as blurred vision and flashing lights in his eyes suggests that the seizure may be related to the treatment he is receiving. The other options, demyelination and hypocalcemia, are possible causes of seizure, but they are less likely to be the cause in this case given the patient's symptoms and the type of medication he is receiving. The patient's age and the fact that he has an untreated HIV infection also make it more likely that the seizure was caused by the medication. Therefore, based on the patient's symptoms and the possible side effects of the medication, it is most likely that the seizure was caused by lactic acidosis. </think> <answer> Lactic acidosis </answer>

